DRIVER-SIDE AIR BAG MODULE [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

id0810b1500500

Purpose

- When the driver-side air bag module receives an impact from a frontal or frontal offset collision, the operation (deployment) of the air bag mediates the impact to the head and face of the driver.

Function

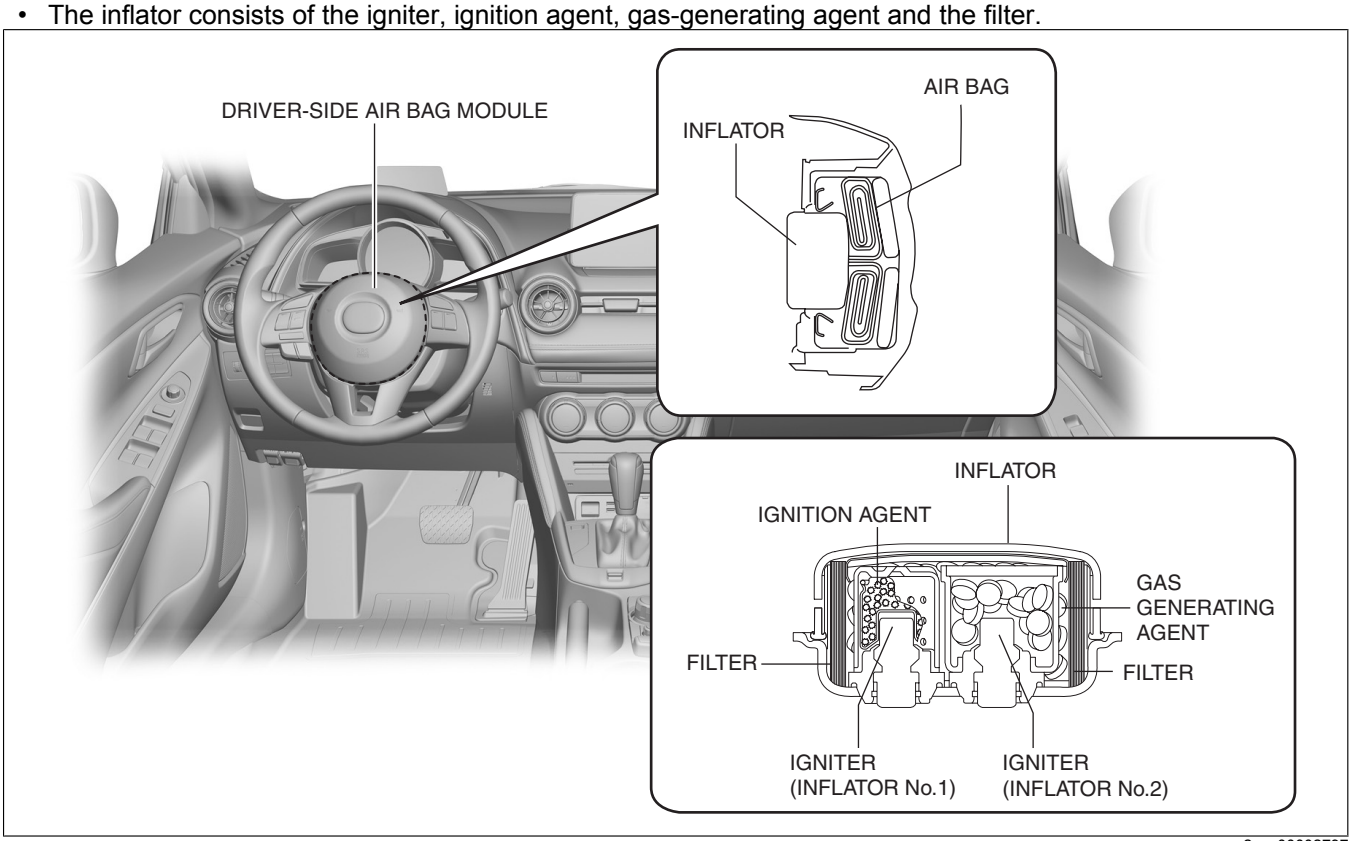
- When the driver-side air bag module receives an impact from a frontal or frontal offset collision, the air bag is operated (deployed) by the operation signal sent from the SAS control module.



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Construction

- The driver-side air bag module is located in the center of the steering wheel.
- The driver-side air bag module consists of the inflator and air bag.
- The inflator consists of the igniter, ignition agent, gas-generating agent and the filter.



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Operation

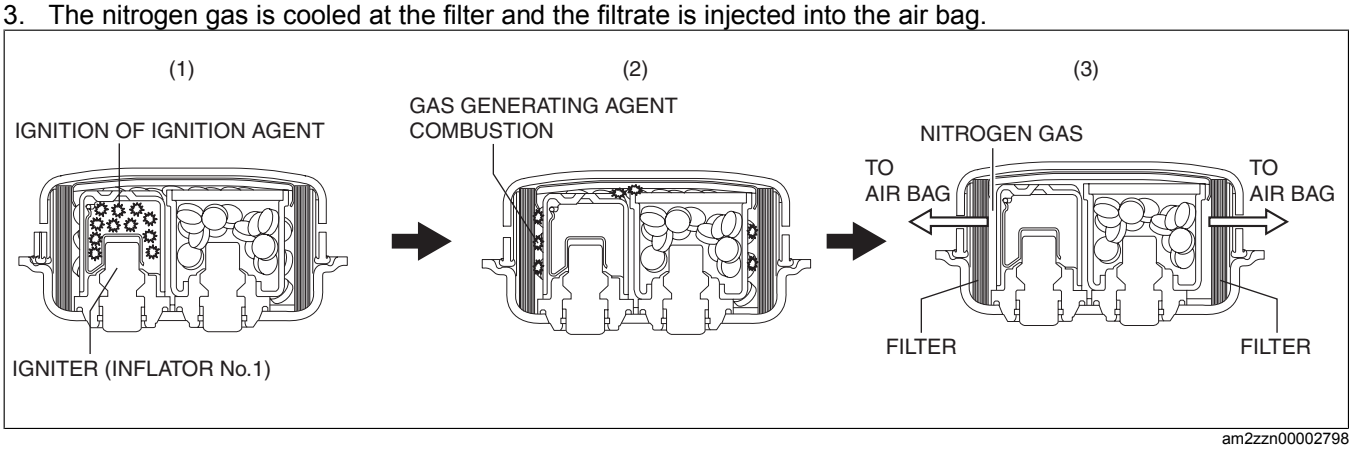
Inflator operation

- The driver-side air bag module operates (deploys) the air bag by activating the internal inflator.
- The inflator operates (deploys) in the following order:

Inflator No.1

1. When the driver-side air bag module receives an operation (deployment) signal from the SAS control module, the igniter built into inflator No.1 builds up heat and ignites the inflation agent.
2. The ignition of the inflation agent causes the combustion of a gas-generating agent which releases nitrogen gas.

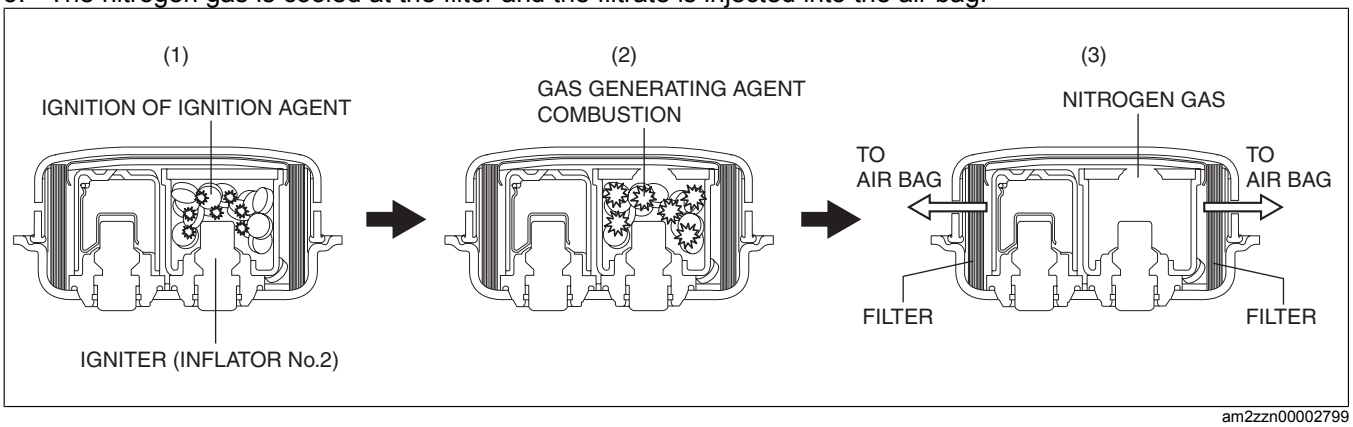
3. The nitrogen gas is cooled at the filter and the filtrate is injected into the air bag.



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Inflator No.2

1. When the driver-side air bag module receives an operation (deployment) signal from the SAS control module, the igniter built into inflator No.2 builds up heat and ignites the inflation agent.
2. The ignition of the inflation agent causes the combustion of a gas-generating agent which releases nitrogen gas.
3. The nitrogen gas is cooled at the filter and the filtrate is injected into the air bag.



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Fail-safe

- Not applicable